

Atoms and isotopes

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 5 starred (★) questions first.

	QUESTION	ANSWER	✓
1	State the approximate radius of an atom.	About 0.1 nm (1×10^{-10} m)	☐
2 ★	Describe the structure of an atom.	A positive nucleus of protons and neutrons, surrounded by negative electrons.	☐
3 ★	State how the radius of a nucleus compares with the radius of an atom.	Less than 1/10 000 of the radius of the atom.	☐
4 ★	State where most of the mass of an atom is concentrated.	In the nucleus.	☐
5	State the sign of the overall charge of the nucleus.	Positive	☐
6	State the relative charge of a proton.	+1 (positive)	☐
7	State the relative charge of a neutron.	0	☐
8	State the relative charge of an electron.	-1 (negative)	☐
9	State the relative mass of a proton.	1	☐
10	State the relative mass of a neutron.	1	☐
11	State the relative mass of an electron.	Very small (about 1/2000)	☐
12	Describe the arrangement of electrons in an atom.	In energy levels at different distances from the nucleus.	☐
13 ★	State how an electron can move to a higher energy level.	By absorbing electromagnetic radiation.	☐
14 ★	State how an electron can move to a lower energy level.	By emitting electromagnetic radiation.	☐
15	Name the two types of particle found in the nucleus of an atom.	Protons and neutrons	☐